Dsp ders 1

kitapta 6. bölümdeyiz

Structures for discrete-time-system

1- Direct form I

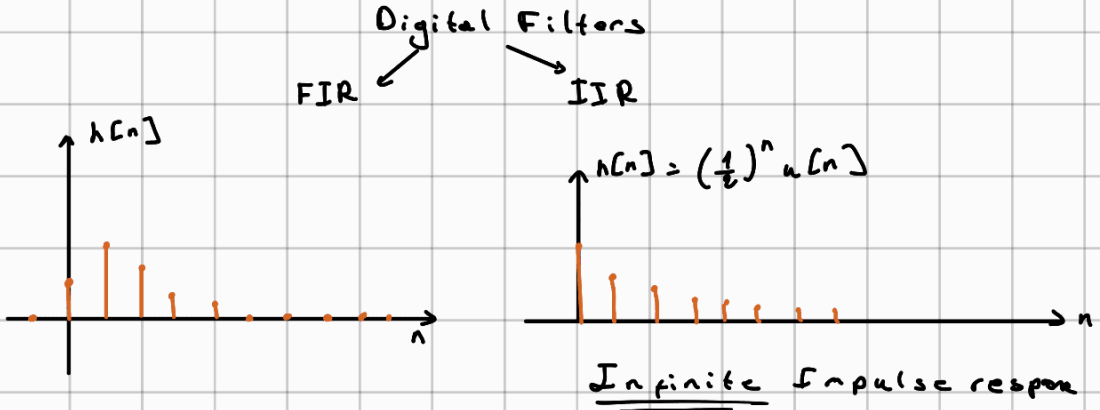
2 - Direct form II

3 - Cascade

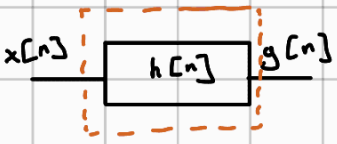
4 - Parallel structures

5 - Transposeel form (flow graph reversal)

Önceden filtreleri ikiye ayırmıştık



Infinite impulse respon



$y[n] = x[n] * h[n]$ → $y[n] = x[n] * h[n]$ ← IIR

FIR

bilgisayarla yapılabilir

bilgilisayarda yapabilmek için belli bir noktadan sonra kesmek gerekiyor ozaman aslında FIR olur böylece yaklaşıklık yapılır.

Ama biz tam olarak bulmak istiyoruz, ozaman başka bir yöntem kullanacağız.

IIR Filter Design

1- specs. (Teknik özellikler)

2- h[n]

3 - realize (gerçeklemek) (this week)

4 - implementation (gerçekleştirmek)

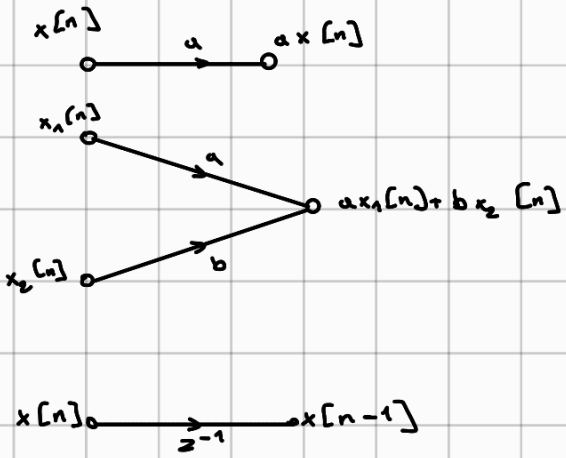
$$y[n] - \sum_{k=1}^{r} a_k y[n-k] = \sum_{k=0}^{q} b_k x[n-k]$$

a_k, b_k real

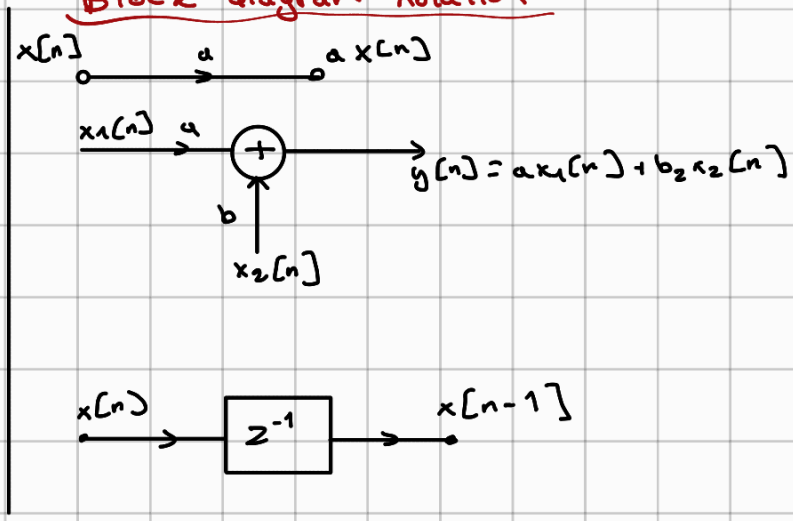
$$H(z) = \frac{\sum_{k=0}^{q} b_k z^{-k}}{1 - \sum_{k=1}^{r} a_k z^{-k}} \quad \text{rational}$$

Özel sadeleşme durumu sağlamazsak bu IIR bir filtrenin sistem fonksiyonu

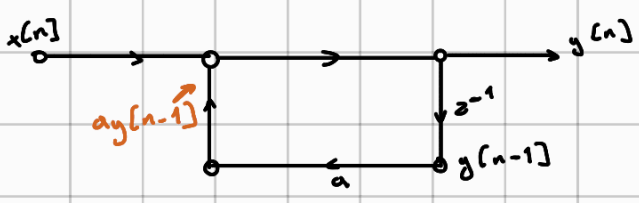
Flow notation

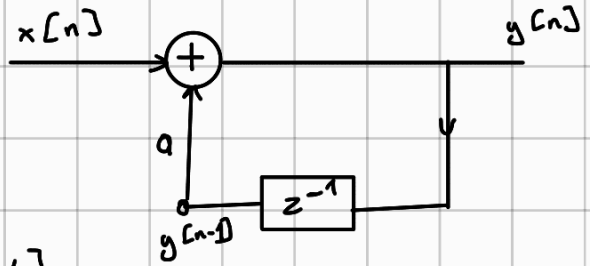


Block diagram notation



$$y[n] = a\, y[n-1] + x[n]$$





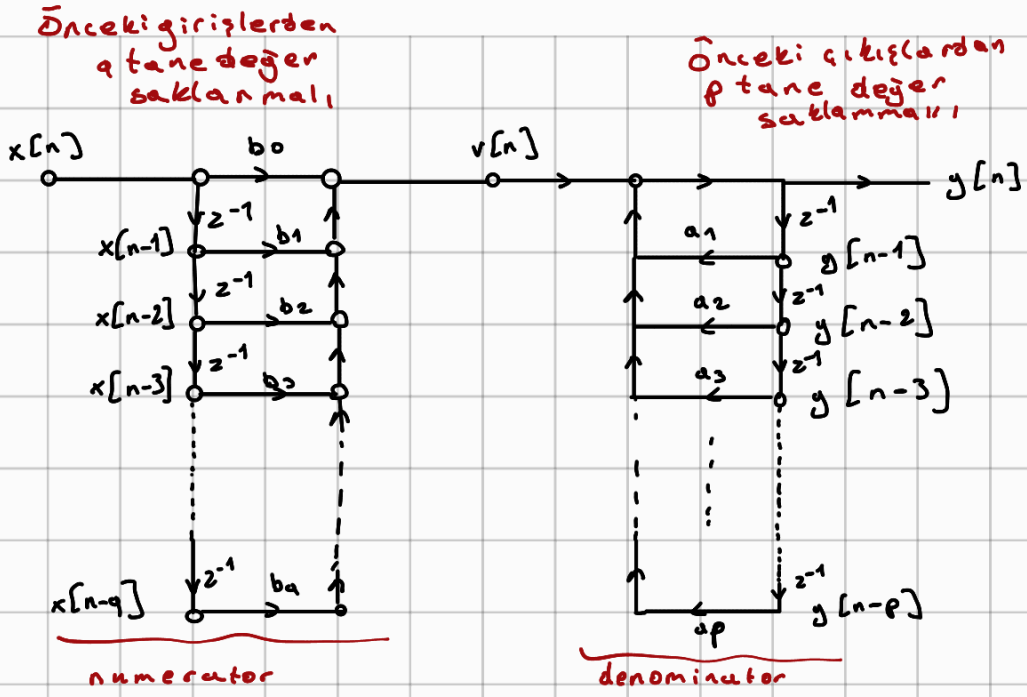
$$y[n] = \sum_{k=1}^{p} a_k y[n-k] + \underbrace{\sum_{k=0}^{q} b_k x[n-k]}_{v[n]}$$

Direct Form-I

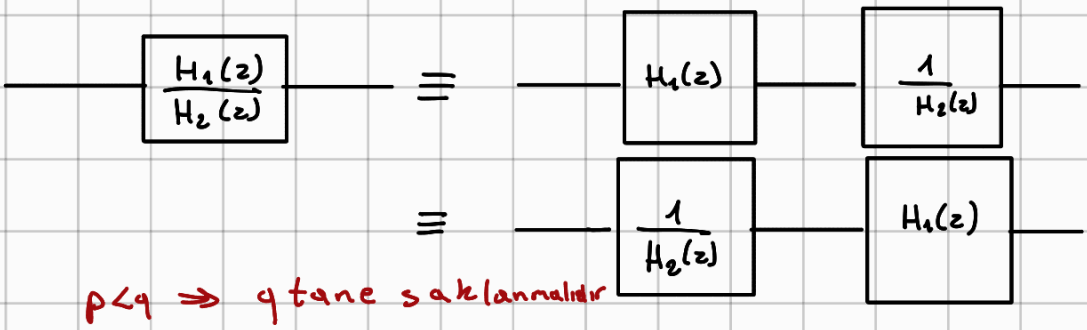
$$y[n] = \sum_{k=1}^{p} a_k y[n-k] + v[n]$$

$$= a_1 y[n-1] + \cdots + a_p y[n-p] + v[n]$$

$$H(z) = \frac{\sum_{k=0}^{q} b_k z^{-k}}{1 - \sum_{k=1}^{r} a_k z^{-k}} \quad \text{rational}$$

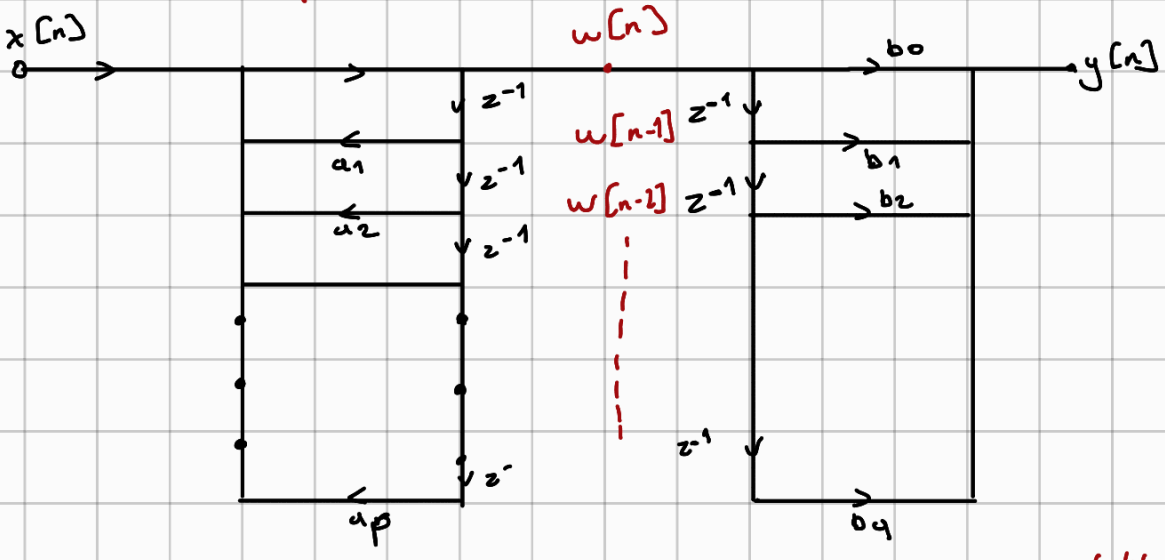


$$H(z) = \frac{H_1(z)}{H_2(z)} = (H_1(z)) \cdot \frac{1}{(H_2(z))}$$



p<q ⇒ q tane saklanmalıdır

hafızada



$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 - a_1 z^{-1} - a_2 z^{-2}}$$

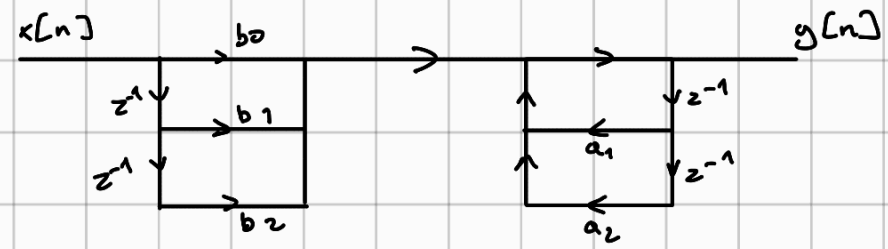
arti arti diye yaptıklarımız artık eksi olacak alarda sola geçiriken - oluyor

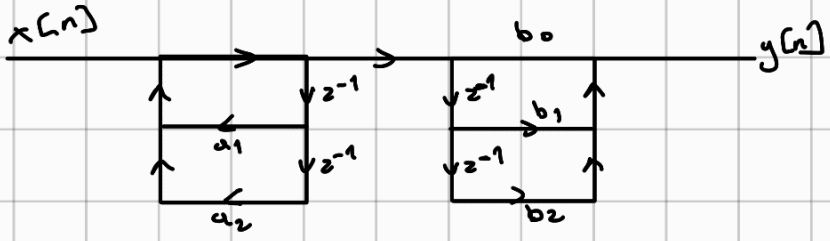
$$y[n] = a_1 y[n-1] + a_2 y[n-2] + b_0 x[n] + b_1 x[n-1] + b_2 x[n-2]$$

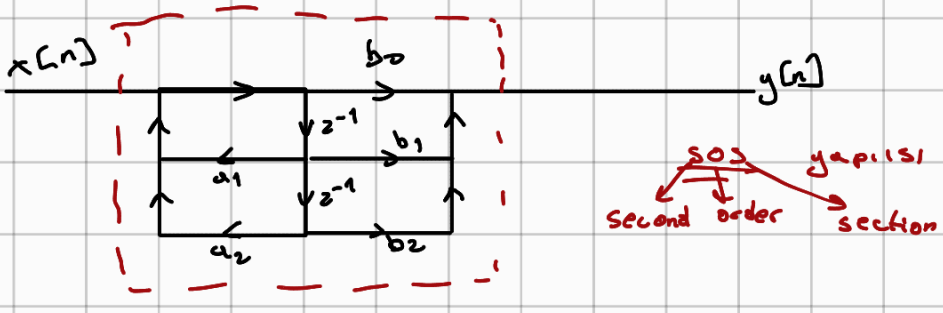
$$y[n] - a_1 y[n-1] - a_2 y[n-2] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2]$$

önceden bunlar arti idi genel gösterimde

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 - a_1 z^{-1} - a_2 z^{-2}}$$

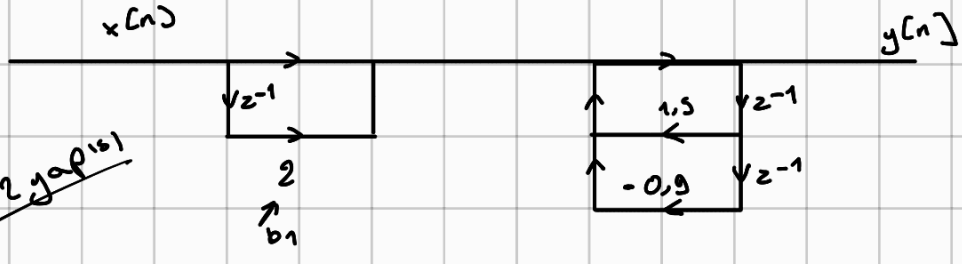




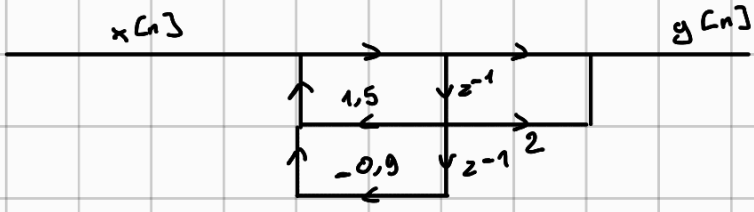


Örnek: $H(z) = \dfrac{1 + 2z^{-1}}{1 - 1{,}5\,z^{-1} + 0{,}9\,z^{-2}}$

Form 1 yapısı



Form 2 yapısı



Cascade Structures

$$H(z) = A = \frac{\prod_k (1 - e_k z^{-1}) \prod_k (1 - f_k z^{-1})(1 - f_k^* z^{-1})}{\prod_k (1 - c_k z^{-1}) \prod_k (1 - d_k z^{-1})(1 - d_k^* z^{-1})}$$

Reel sıfırlar

Reel kutuplar

$$(1 - d_k z^{-1})(1 - d_k^* z^{-1}) = 1 - 2\,\mathrm{Re}[d_k] z^{-1} + |d_k|^2 z^{-2}$$

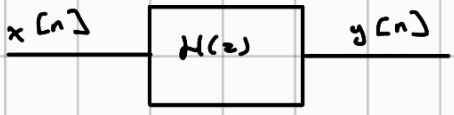
$$= 1 + \beta_{1k} z^{-1} + \beta_{2k} z^{-2}$$

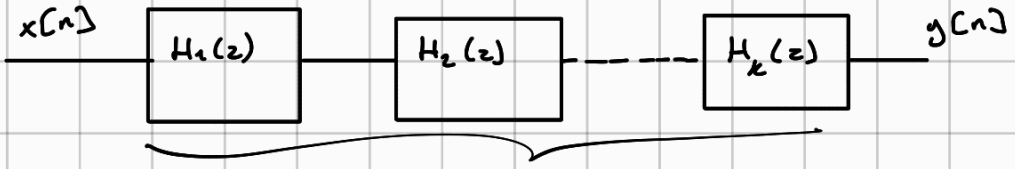
O zaman $H(z)$ tekrar bu şekilde yazabiliriz

$$H(z) = A \prod_k H_k(z)$$

$$H_k(z) = \frac{1 + \beta_{1k} z^{-1} + \beta_{2k} z^{-2}}{1 - \alpha_{1k} z^{-1} - \alpha_{2k} z^{-2}}$$

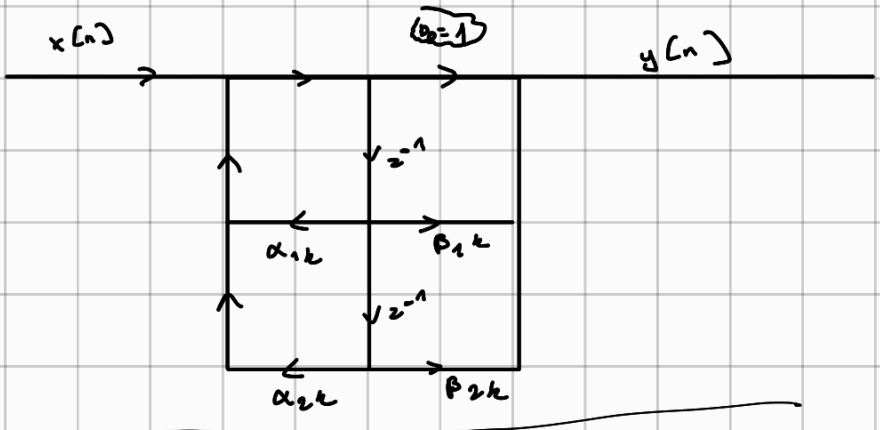
s.o.s → section
second → order





Cascade

SOS Form

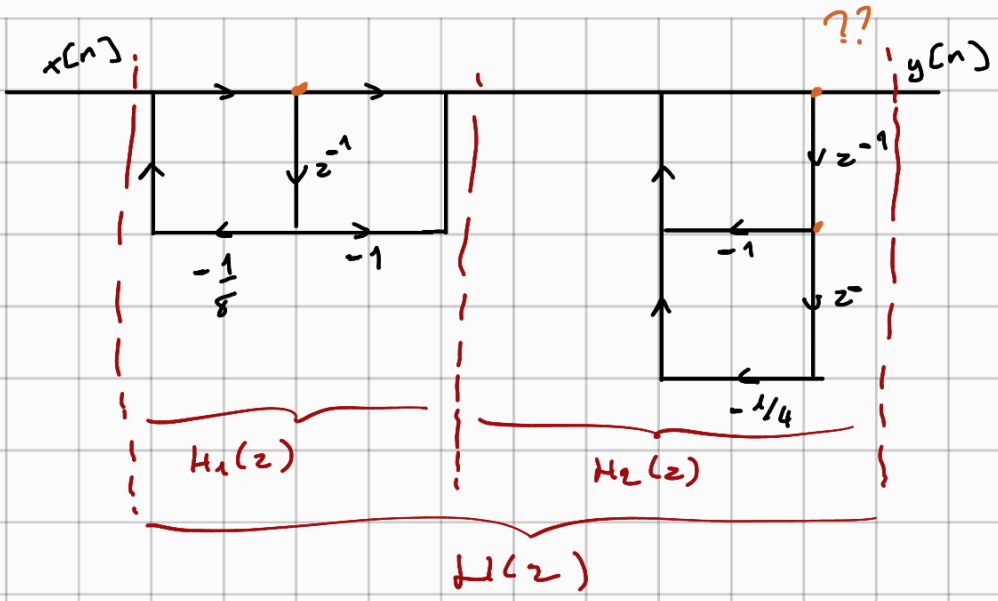


$H_k(z)$

Örnek:

$$H(z) = \frac{1 - z^{-1}}{(1 + \frac{1}{8} z^{-1})(1 + z^{-1} + \frac{1}{4} z^{-2})} = \left(\frac{1}{1 + \frac{1}{8} z^{-1}}\right)\left(\frac{1 - z^{-1}}{1 + z^{-1} + \frac{1}{4} z^{-2}}\right)$$

$$= \left(\frac{1 - z^{-1}}{1 + \frac{1}{8} z^{-1}}\right)\left(\frac{1}{1 + z^{-1} + \frac{1}{4} z^{-2}}\right)$$



Burda 3 tane değeri hafızada saklamak gerekir.
(gecikmelerden dolayı)

Paralleled implementation

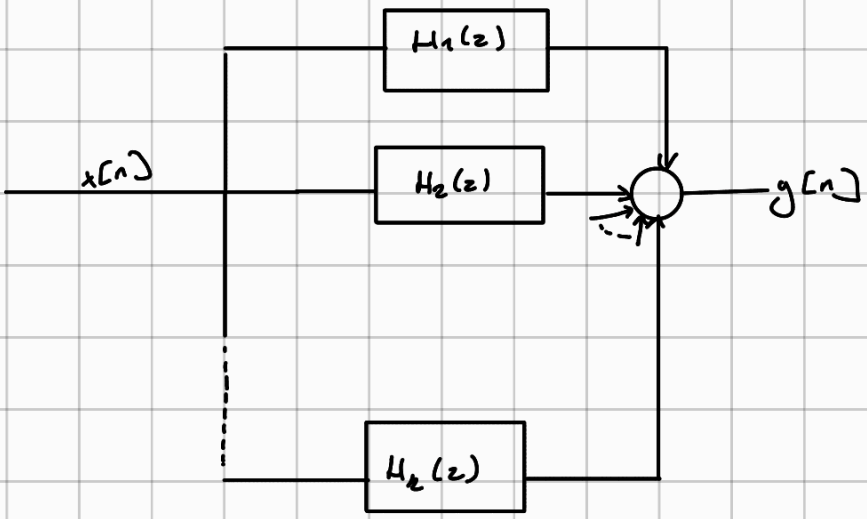
S.O.S ların toplamı şeklinde ifade edilmelidir.

$$H(z) = \frac{\sum_{k=1}^{p} b_k z^{-k}}{1 - \sum_{k=1}^{q} a_k z^{-k}}$$

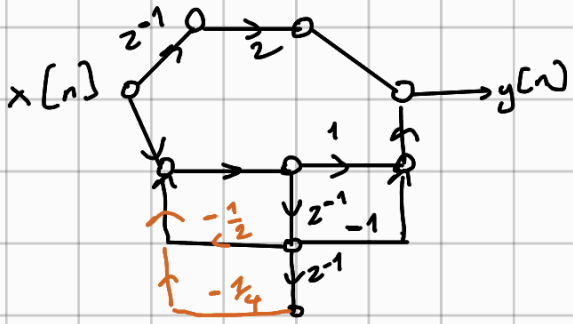
$$= \sum_k A_k z^{-k} + \sum_k \frac{B_k}{1 - g_k z^{-1}} + \sum_k \frac{C_k + D_k z^{-1}}{1 - h_{1k} z^{-1} - k_{2k} z^{-2}}$$

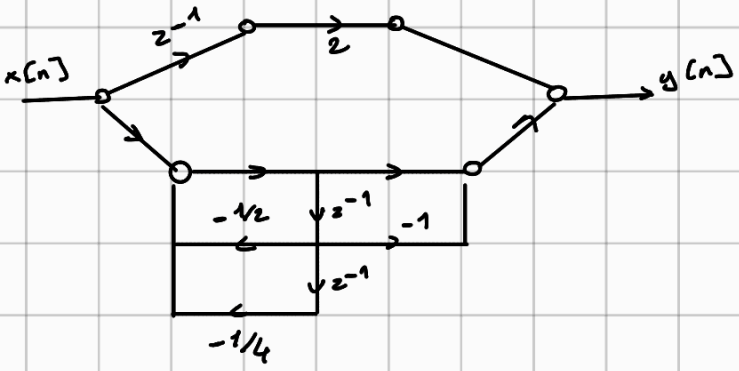
en genel hal

$$= \sum_k H_k(z)$$



$$H(z) = \underbrace{2z^{-1}}_{H_1(z)} + \underbrace{\frac{1 - z^{-1}}{1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}}_{H_2(z)}$$

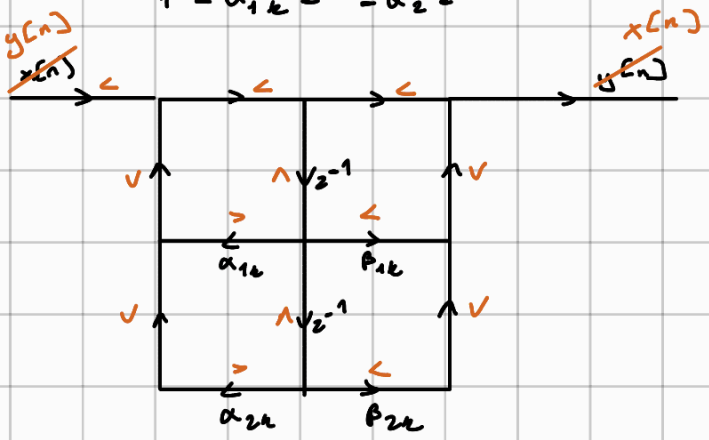




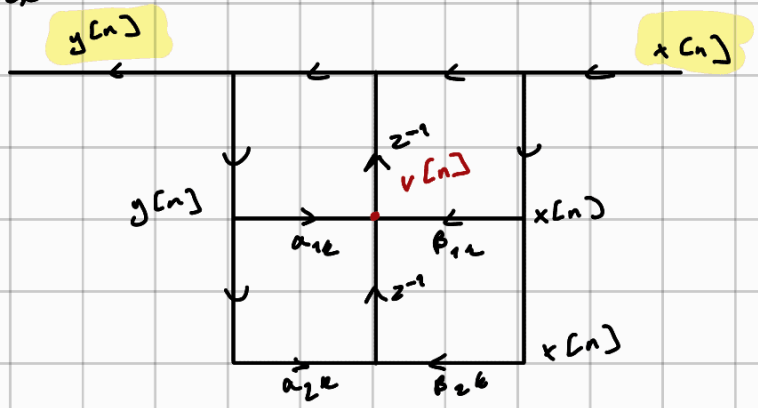
Transpozition theorem
(Flow-graph reversal theorem)

Teoremegöre x[n] ve y[n] 'in yeri değiştirilip tümokların yönü terslenirse yenisisten eskinin eşidir.

$$H_k(z) = \frac{1 + \beta_{1k} z^{-1} + \beta_{2k} z^{-2}}{1 - \alpha_{1k} z^{-1} - \alpha_2 z^{-2}}$$



yeni sisteni tekrar çizelim



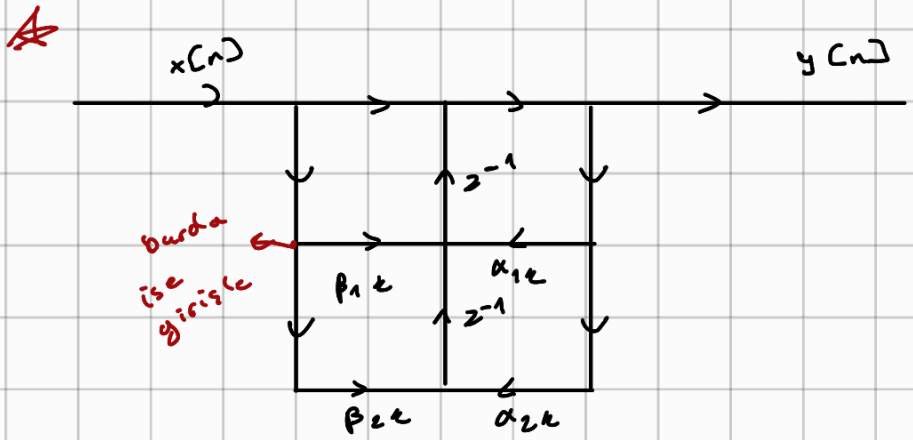
$$v[n] = x[n] \cdot \beta_{1k} + y[n]\,\alpha_{1k} + y[n-1]\,\alpha_{2k} + x[n-1]\,\beta_{2k}$$

$$y[n] = x[n] + v[n-1]$$

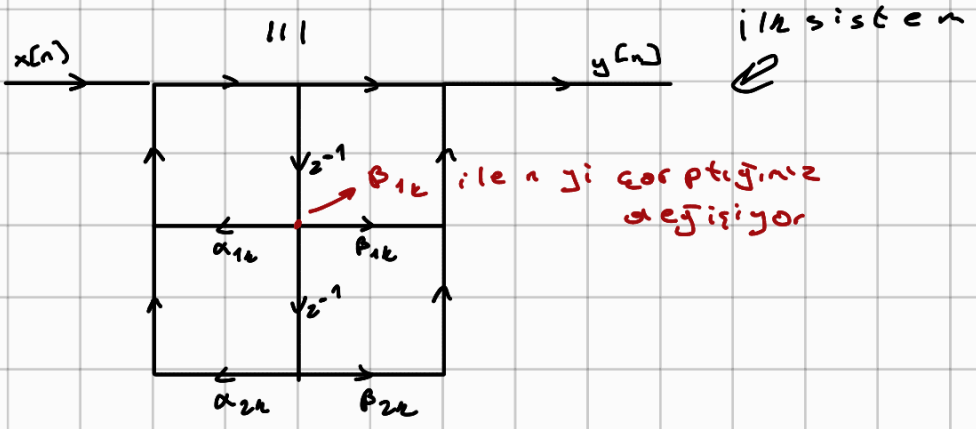
$$= x[n] + x[n-1]\,\beta_{1k} + x[n-2]\,\beta_{2k} + y[n-1]\,\alpha_{1k} + y[n-2]\,\alpha_{2k}$$

$$\frac{y(z)}{X(z)} = \frac{1 + \beta_{1k} z^{-1} + \beta_{2k} z^{-2}}{1 - \alpha_{1k} z^{-1} - \alpha_2 z^{-2}}$$

Simetriğini alırsak bizim alıştığımız formatta



burda ise işaretler değişiyor



ilk sistem

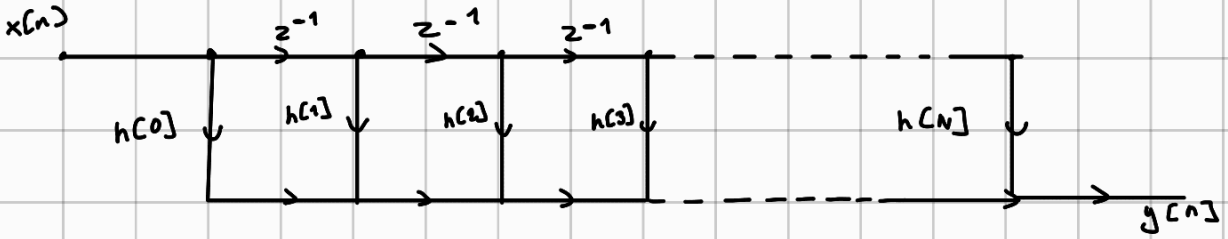
β_{1k} ile a ji çarptığınız değişiyor

FIR realization

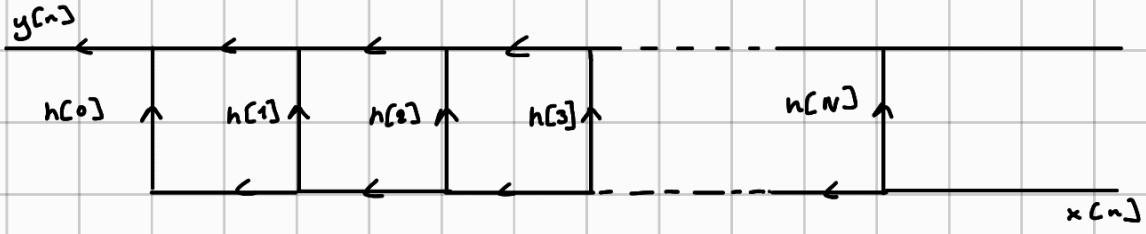
$h[n]$: finite

$$H(z) = \sum_{k=0}^{\infty} h[k] z^{-k}$$

$$\Rightarrow y[n] = \sum_{k=0}^{n} h[k]\, x[n-k] = h[0]x[n] + h[1]x[n-1] + h[2]x[n-2] + \cdots + h[n]x[n-N]$$

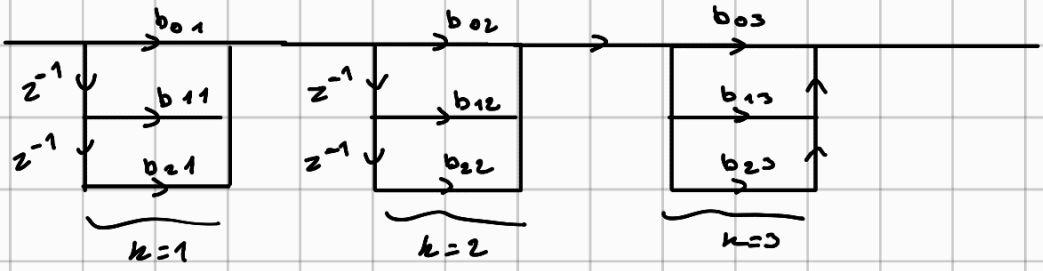


use transposition theorem:



In cascade Structure:

$$H(z) = \sum_{k=0}^{N} h[k]\, z^{-k} = \prod_{k=1}^{N} (b_{0k} + b_{1k} z^{-1} + b_{2k} z^{-2})$$

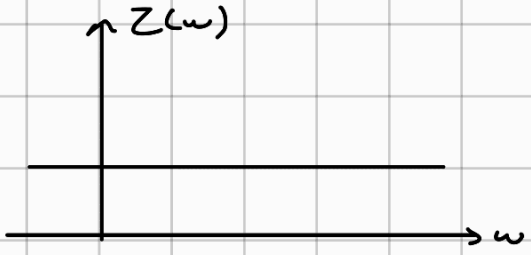


$H(e^{jw})$ $\quad |H(e^{jw})|$ $\quad \angle H(e^{jw})$

$$\tau(w) = -\frac{d \angle H(e^{jw})}{dw}$$

group delay

$\tau(w)$ = constant



Önceden böyle bir örnek göstermiştik



0,2 Hz
0,4 Hz
0,8 Hz

$\angle H(e^{jw}) = \alpha w + \beta$ yapısında ise $\tau(w)$ = constant

Linear Phase Filter

IIR X (hiç bir zaman sağlayamaz)

FIR sağlayabilir

Type-I

$h[n] = h[M-n]$ $\quad 0 \le n \le M$

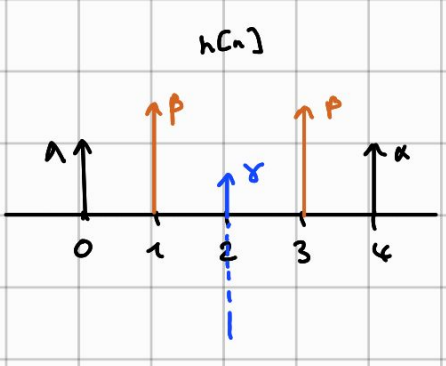
M : even integer (çift)

$M=4 \Rightarrow 0 \le n \le 4$

$h[0] = h[4]$

$h[1] = h[3]$

$h[2]$



Type II

$h[n] = h[M-n]$ $\quad 0 \le n \le M$

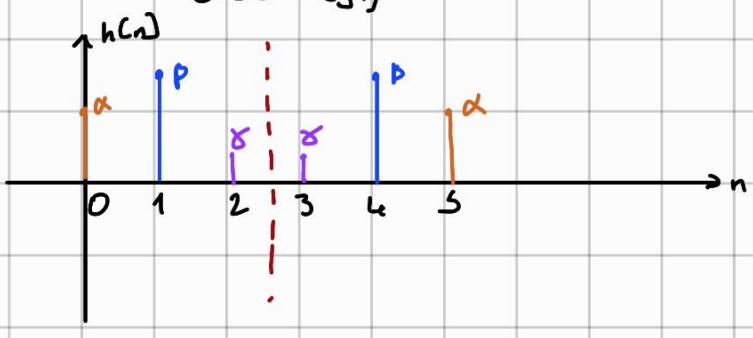
M: odd integer (tek sayı)

$M=5 \Rightarrow 0 \le n \le 5$

$h[0] = h[5]$

$h[1] = h[4]$

$h[2] = h[3]$

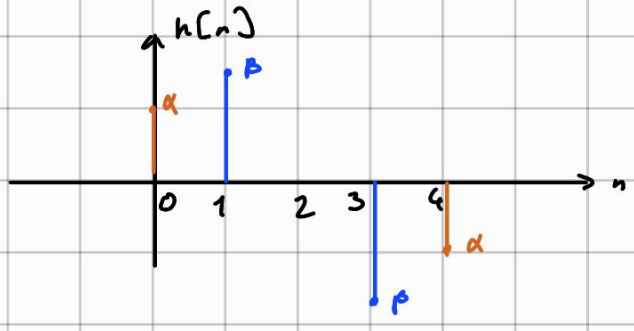


Type III

$h[n] = -h[M-n]$ $\quad 0 \le n \le M$

M: even integer (çift sayı)

$M=4$



Type IV

$h[n] = -h[M-n]$ $\quad 0 \le n \le M$ $\quad$ M : odd integer

$M=5$

